

SIMATS ENGINEERING

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Machine Learning

Experiment -17

Implement Mobile Price Prediction using appropriate machine learning algorithm

Code:

```
# Play Tennis Prediction using Random Forest Classifier

import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
from sklearn.preprocessing import LabelEncoder

# Load dataset
df = pd.read_csv("play_tennis (1).csv")

# Display dataset
print("Dataset:")
print(df.head())

# Convert categorical columns into numerical values
# This includes features and the target 'play'
for column in df.select_dtypes(include=["object"]).columns:
    encoder = LabelEncoder()
    df[column] = encoder.fit_transform(df[column])

# Separate features and target
X = df.drop("play", axis=1) # Corrected column name
y = df["play"] # Corrected column name

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

# Create Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

# Train the model
model.fit(X_train, y_train)

# Predict 'play' outcomes
y_pred = model.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)

print("\nPlay Tennis Prediction") # Updated title
print("-----")
```

```

print("Accuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
# Use 'Yes' and 'No' for target names if original labels are desired for report
# For simplicity here, using the numerical labels (0 and 1) directly
print(classification_report(y_test, y_pred))

# Actual vs Predicted
result = pd.DataFrame({
    "Actual Play": y_test.values, # Updated column name
    "Predicted Play": y_pred # Updated column name
})

print("\nActual vs Predicted Play:") # Updated title
print(result)

```

OUTPUT:

```

*** Dataset:
   day outlook temp humidity wind play
0 D1   Sunny   Hot     High   Weak   No
1 D2   Sunny   Hot     High  Strong   No
2 D3  Overcast  Hot     High   Weak   Yes
3 D4    Rain  Mild     High   Weak   Yes
4 D5    Rain  Cool    Normal   Weak   Yes

Play Tennis Prediction
-----
Accuracy: 66.67%

Confusion Matrix:
[[0 1]
 [0 2]]

Classification Report:

```

	precision	recall	f1-score	support
0	0.00	0.00	0.00	1
1	0.67	1.00	0.80	2
accuracy			0.67	3
macro avg	0.33	0.50	0.40	3
weighted avg	0.44	0.67	0.53	3

```


Actual vs Predicted Play:
   Actual Play Predicted Play
0          0          1
1          1          1
2          1          1

```

